



Clinic Management System

A Relational Database Design & Implementation Project

Team Members

Yousef elsayed Mohamed · 221001880

Abdelrahman nasr · 222000063

Instructor: Dr. Mohamed Elsayeh

Project Overview

Purpose

The Clinic Management System is designed to manage patients, appointments, clinics, doctors, and departments. It simplifies scheduling, stores patient records, and tracks medical history — all backed by a fully relational SQL database.

Project Goals

- Design a normalized relational database for clinical operations
- Enable efficient CRUD operations (Create, Read, Update, Delete)
- Write practical SQL queries for healthcare use cases
- Maintain data integrity using foreign keys & constraints

Department

10 departments including Cardiology, Neurology, Oncology...

Clinic

10 clinics linked to their respective departments

Doctor

10 doctors assigned to departments

Patient

10 patients with personal & contact information

Appointment

10 appointments with diagnosis, cost, status & timing

Database Schema

- Department

<u>Department ID</u>	Name
----------------------	------

- Clinic

<u>Clinic ID</u>	Department ID	Name	Address
------------------	---------------	------	---------

- Doctor

<u>Doctor ID</u>	Name	Phone no.	Address	Department ID
------------------	------	-----------	---------	---------------

- Patient

<u>Patient ID</u>	Name	Phone no.	Address	B.Date	job
-------------------	------	-----------	---------	--------	-----

- Appointment

<u>APP. ID</u>	Date	Patient ID	Doctor ID	start time
End time	Cost	schedule	in progress	postponed
Diagnosis				

Department

DepartmentID (PK), Name

Clinic

ClinicID (PK), Name, Address, DepartmentID (FK)

Doctor

DoctorID (PK), Name, PhoneNumber, Address, DepartmentID (FK)

Patient

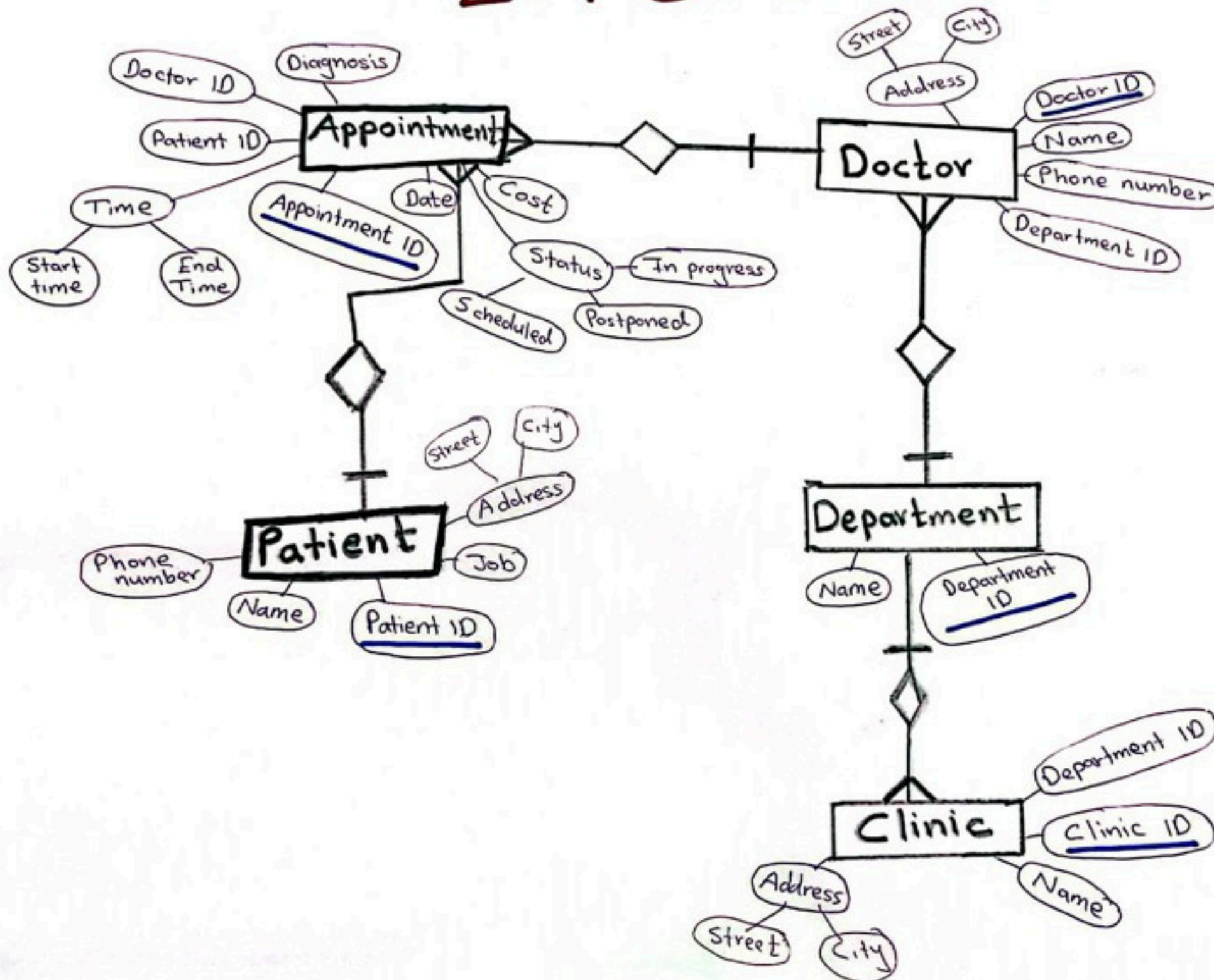
PatientID (PK), Name, PhoneNumber, Address, BirthDate, Job

Appointment

AppointmentID (PK), Date, PatientID (FK), DoctorID (FK), StartTime, EndTime, Cost, Status, Diagnosis

ER Diagram

ER DIAGRAM



Relationships

1 : N

Department → Clinics

One department manages many clinics

1 : N

Department → Doctors

One department contains many doctors

1 : N

Doctor → Appointments

One doctor handles many appointments

1 : N

Patient → Appointments

One patient can have multiple appointments

N : 1

Clinic → Department

Each clinic belongs to one department

Sample SQL Queries

Query 1 – Patients diagnosed with Fatty Liver in the last year

```
125      -- 1. Patients diagnosed with 'Fatty liver' in the last year
126 •    SELECT Name
127      FROM Patient
128  ⊖    WHERE PatientID IN (
129          SELECT PatientID
130          FROM Appointment
131          WHERE Diagnosis = 'Fatty liver'
132          AND Date >= DATE_SUB(CURDATE(), INTERVAL 1 YEAR)
```

Sample SQL Queries

Query 2 – Addresses of Cardiology Clinics (JOIN)

```
132         AND Date >= DATE_SUB(CURDATE(), INTERVAL 1 YEAR)
133     );
134
135     -- 2. Addresses of cardiology clinics
136 • SELECT Clinic.Address
137     FROM Clinic
138     JOIN Department ON Clinic.DepartmentID = Department.DepartmentID
139     WHERE Department.Name = 'Cardiology';
140
```

Sample SQL Queries

Query 3 – Total Amount Paid by Patient #12527 in the Last 3 Years (Aggregate)

```
139 WHERE Department.Name = 'Cardiology';
140
141 -- 3. Total money paid by patient with ID 12527 in the last 3 years
142 • SELECT SUM(Cost) AS TotalPaid
143 FROM Appointment
144 WHERE PatientID = 12527
145     AND Date >= DATE_SUB(CURDATE(), INTERVAL 3 YEAR);
146
```

Data Population & CRUD Operations

Sample Data Overview

Table	Records	Key Fields
Department	10	Cardiology, Neurology, Oncology, Pediatrics...
Clinic	10	Heart Clinic, Brain Clinic, Cancer Clinic...
Doctor	10	Dr. Ayman, Dr. Leila, Dr. Sara...
Patient	10	Ali Ahmed, Sara Youssef, Tamer Sherif...

Sample UPDATE Operations

- Update patient address & phone (PatientID 12530)
- Set appointment status to 'Postponed' (AppointmentID 20003)
- Update clinic address (ClinicID 101)

CREATE

INSERT INTO – Add new departments, clinics, doctors, patients, appointments

READ

SELECT – Retrieve patient lists, diagnoses, clinic addresses, cost totals

UPDATE

UPDATE SET – Modify patient contact info, appointment status, clinic addresses

DELETE

DELETE FROM – Remove records while maintaining referential integrity

Key Takeaways

Designed a fully normalized relational database with 5 entities

Implemented foreign key constraints for full data integrity

Wrote complex queries using JOINS, subqueries & aggregates

Populated all tables with 10+ realistic test records each

Demonstrated CRUD operations across all tables

Thank You!

Questions?